

Power of a Point + Radical Axis

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1 Power of a Point Theorem

1) Let A, B, C, D be points in the plane with $P = AB \cap CD$ such that P either lies on both segments AB and CD or neither. Then,

$$ABCD \text{ cyclic} \iff PA \times PB = PC \times PD.$$

2) Let A, B, C be points in the plane with $P \in BC$. Then,

$$PA \text{ tangent to } (ABC) \iff PA^2 = PB \times PC.$$

Prove these theorems.

2 Defining the Power of a Point

The **power** of a point P with respect to a circle ω with center O and radius R is defined as

$$\text{pow}(P, \omega) = OP^2 - R^2$$

When is the power of a point positive, negative, or zero? How does this definition relate to the previous theorem?

3 Radical Axis

What is the locus of points that have equal powers with respect to two given non-concentric circles? This is known as the radical axis.

Spoiler!: The radical axis of two circles is a line perpendicular to the line connecting the centers of the circles.

How can we geometrically construct the radical axis?

4 Radical Center

What is the locus of points that have equal powers with respect to three given non-concentric circles? This is known as the radical center.

Radical Axis Theorem: The pairwise radical axes of three non-concentric circles are concurrent at the radical center.

Use this to prove the existence of the orthocenter.

5 Degenerate Circles

We can think of points as circles with radius 0.

Use the Radical Axis Theorem to prove the existence of the circumcenter.

6 Problems

1. Let Γ_1 and Γ_2 be two intersecting circles. Let a common tangent to Γ_1 and Γ_2 touch Γ_1 at A and Γ_2 at B . Show that the common chord of Γ_1 and Γ_2 , when extended, bisects segment AB .
2. Let C be a point on a semicircle of diameter AB and let D be the midpoint of arc AC . Let E be the projection of D onto the line BC and F the intersection of line AE with the semicircle. Prove that BF bisects the line segment DE .
3. (APMO 2020 P1) Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC . The tangent to Γ at A intersects the parallel line to BA through D at point E . The segment CE intersects Γ again at F . Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.
4. Let ABC be an acute triangle. Let the line through B perpendicular to AC meet the circle with diameter AC at points P and Q , and let the line through C perpendicular to AB meet the circle with diameter AB at points R and S . Prove that P, Q, R, S are concyclic.
5. (2009 USAMO P1) Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .
6. (2012 USAJMO P1) Given a triangle ABC , let P and Q be points on segments $\overline{AB}$ and $\overline{AC}$, respectively, such that $AP = AQ$. Let S and R be distinct points on segment $\overline{BC}$ such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P, Q, R, S are concyclic (in other words, these four points lie on a circle).
7. (Euler's Theorem/Inequality) Let $\triangle ABC$ be a triangle. If O and I are its circumcenter and incenter, respectively, and R and r are its

circumradius and inradius, respectively, then prove that $OI^2 = R(R - 2r)$, and in particular $R \geq 2r$.

8. (Iran TST 2011 P1) In acute triangle ABC angle B is greater than C . Let M is midpoint of BC . D and E are the feet of the altitude from C and B respectively. K and L are midpoint of ME and MD respectively. If KL intersect the line through A parallel to BC in T , prove that $TA = TM$.
9. (Al-Khwarizmi 2024 P7) Two circles with centers O_1 and O_2 intersect at P and Q . Let ω be the circumcircle of the triangle PO_1O_2 ; the circle ω intersect the circles centered at O_1 and O_2 at points A and B , respectively. The point Q is inside triangle PAB and PQ intersects ω at M . The point E on ω is such that $PQ = QE$. Let ME and AB meet at L , prove that $\angle QLA = \angle MLA$.
10. (BMO 2015 P2) Let $\triangle ABC$ be a scalene triangle with incentre I and circumcircle ω . Lines AI, BI, CI intersect ω for the second time at points D, E, F , respectively. The parallel lines from I to the sides BC, AC, AB intersect EF, DF, DE at points K, L, M , respectively. Prove that the points K, L, M are collinear.